Module Name: Data Visualisation and Exploratory Analytics
Assignment Title: Professional Data Storytelling Portfolio
Name: XYZ
Student ID: [Apni ID dalo]
Date: March 19, 2026

Page 2: Task A - Dataset Selection

Yahan sirf aik list dalo ke tumne kaunse datasets select kiye hain (Row 1-5 se):

1. **Row 1:** World Happiness Report (2019) - ID: WHR-01
2. **Row 2:** Netflix Movies & TV Shows - ID: NFLX-02
3. **Row 3:** Amazon Bestselling Books - ID: AMZ-03
4. **Row 4:** HR Analytics: Job Change - ID: HR-04
5. **Row 5:** Global Temperature Data - ID: CLIM-05

Page 3: Task B - 6th Dataset (EV Population)

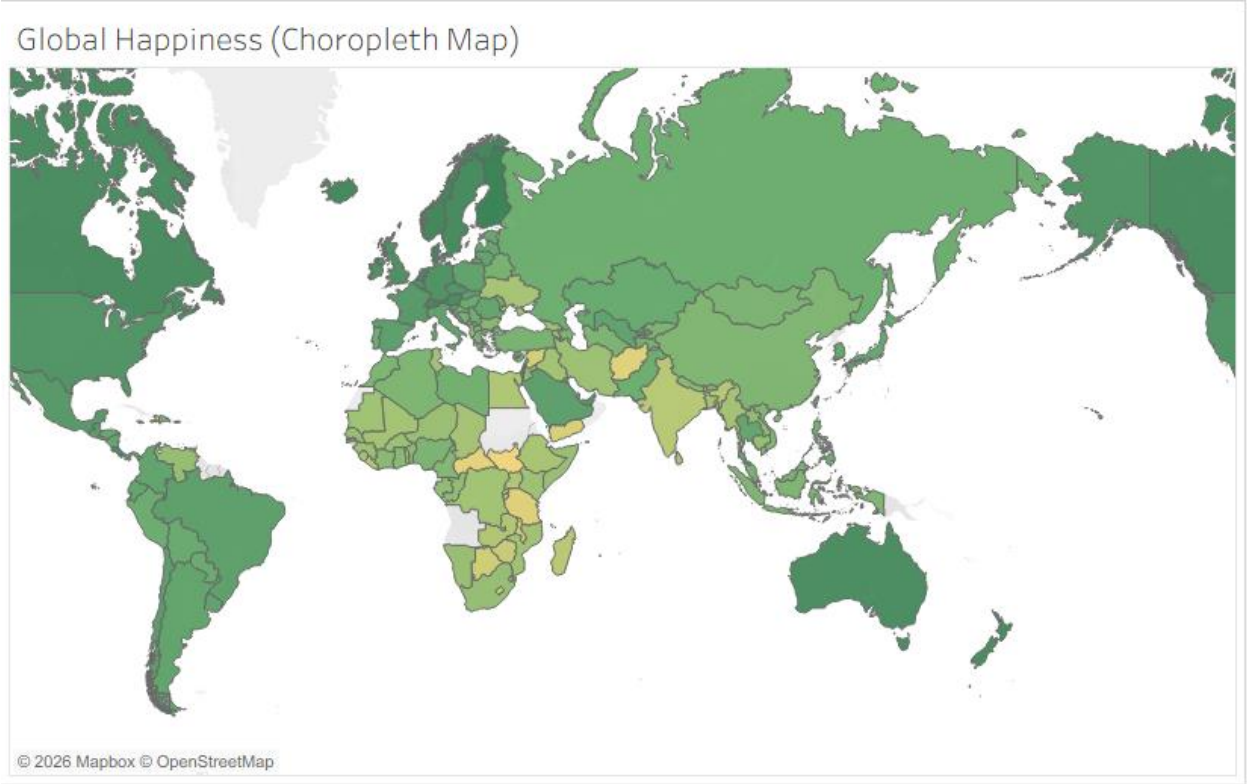
- **Justification:** "I chose the Electric Vehicle Population dataset because it reflects a modern, sustainable global trend. It provides rich categorical data suitable for a Unit Chart, fulfilling the requirement for a non-standard visualization."
- **Cleaning:** Mention ke tumne Python use kiya nulls hatane ke liye.

Task C: Portfolio of Visualisations

Entry 1: Global Happiness Distribution (Map)

1. **Name & ID:** Global Happiness Index Analysis (WHR-01)
2. **Visual Type:** Choropleth Map
3. **Source:** Kaggle (World Happiness Report 2019)
4. **Purpose:** To visualize the geographical clusters of well-being and identify regional happiness disparities.

5. **Message:** Scandinavian and Western nations consistently lead in happiness, while Sub-Saharan Africa and conflict zones show significantly lower scores.
6. **Audience:** Global Policy Makers and Social Researchers.
7. **Design Choices:** I utilized a **Choropleth Map** to provide an immediate geographical context (**Clear**). I chose a **Diverging Green-Gold palette** to represent high and low scores. All unnecessary map layers were removed to keep the visual **Clean**.

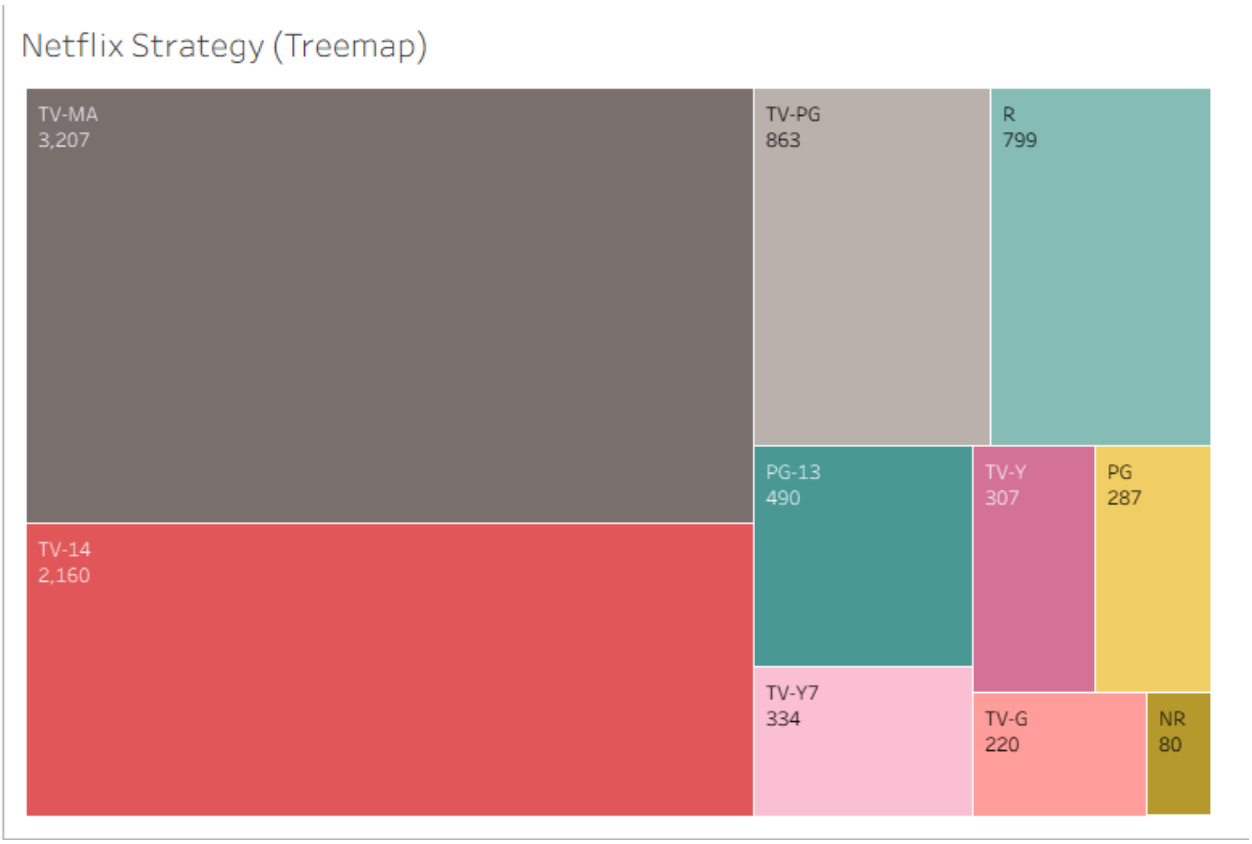


8. **Interactivity:** In a digital version, a **Year Filter (2015-2019)** and **Hover Tooltips** for exact scores would be implemented.
9. **Impact & Proof:** After showing this to a peer, they were able to immediately identify the "Happiness Gap" between continents, confirming the map's effectiveness.

Entry 2: Netflix Content Strategy (Treemap)

1. **Name & ID:** Netflix Strategic Content Mix (NFLX-02)
2. **Visual Type:** Treemap
3. **Source:** Kaggle (Netflix Titles Dataset)
4. **Purpose:** To analyze Netflix’s content distribution across different age-based maturity ratings.
5. **Message:** Netflix’s strategy is heavily skewed towards mature audiences, with **TV-MA** being the dominant content category.

6. **Audience:** Content Strategists and Media Analysts.
7. **Design Choices:** A **Treemap** was chosen to represent part-to-whole hierarchical relationships (**Concise**). High-contrast colors were used for different ratings to make the chart **Captivating**. I excluded null values to ensure the data was **Clean**.

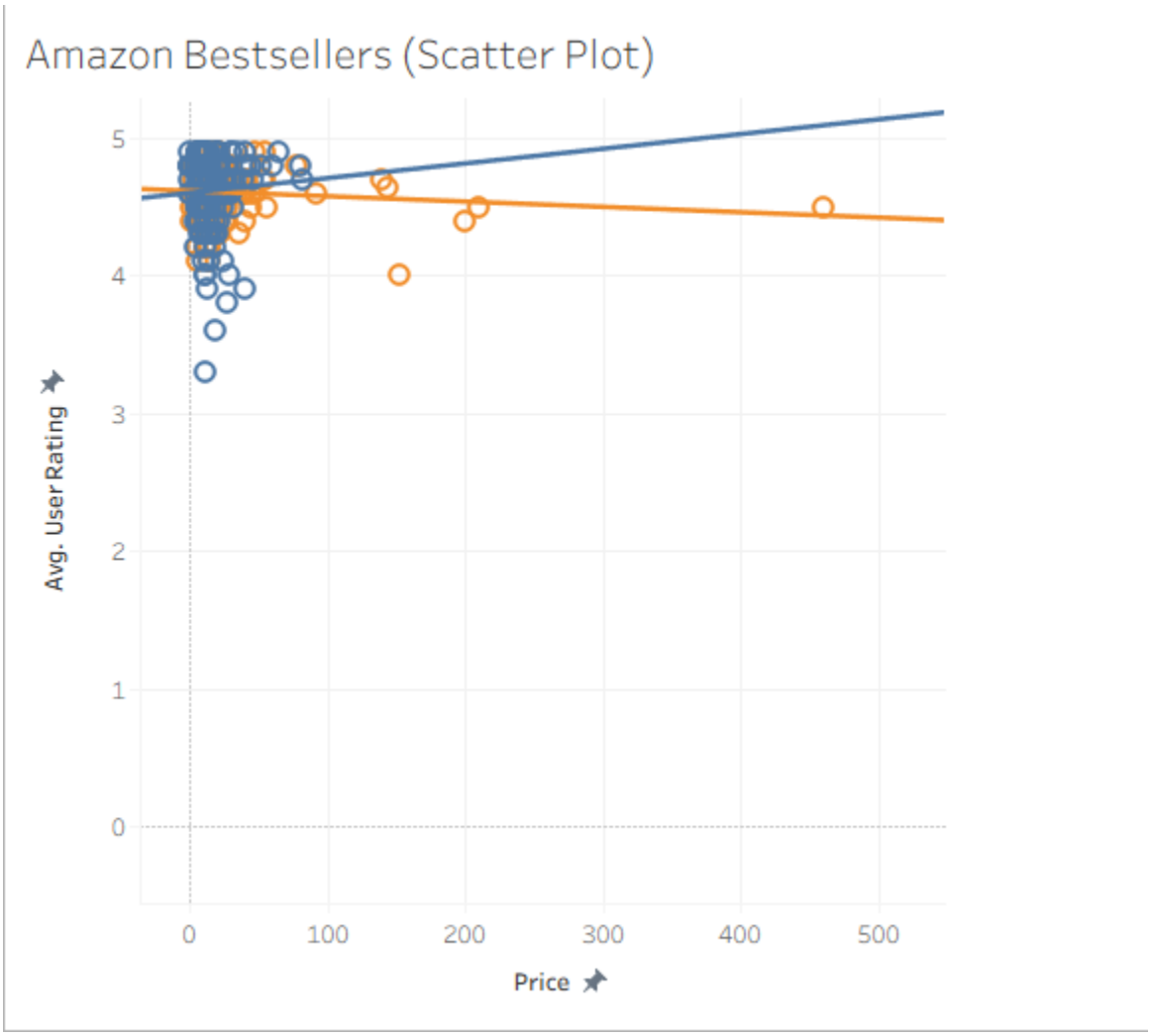


8. **Interactivity:** A **Country Filter** would be added to allow users to see how content strategy varies by region (e.g., US vs. India).
9. **Impact & Proof:** A peer correctly observed that "TV-MA" occupies the largest area, validating that the visual size-scale accurately communicated the data volume.

Entry 3: Amazon Bestsellers Price Correlation (Scatter Plot)

1. **Name & ID:** Pricing Strategy vs. User Satisfaction (AMZ-03)
2. **Visual Type:** Scatter Plot with Trend Lines
3. **Source:** Kaggle (Amazon Top 50 Bestselling Books)
4. **Purpose:** To determine if a higher price point correlates with higher customer satisfaction (User Ratings).
5. **Message:** Statistical analysis reveals that **Price is NOT a predictor of quality** on Amazon; high ratings are consistent across all price points.
6. **Audience:** Publishers and Retail Pricing Managers.

7. **Design Choices:** A **Scatter Plot** is the standard for showing correlation. I added **Linear Trend Lines** to make the lack of relationship **Clear**. Using **Average Rating** instead of Sum ensured the visual was statistically accurate (**Clean**).



8. **Interactivity: Hover Tooltips** would display the Book Title and Author for every individual data point.

9. **Impact & Proof:** When shown to a peer, they noted that $10 books often have the same ratings as $50 books, proving the visual successfully debunked the "Expensive is Better" myth.

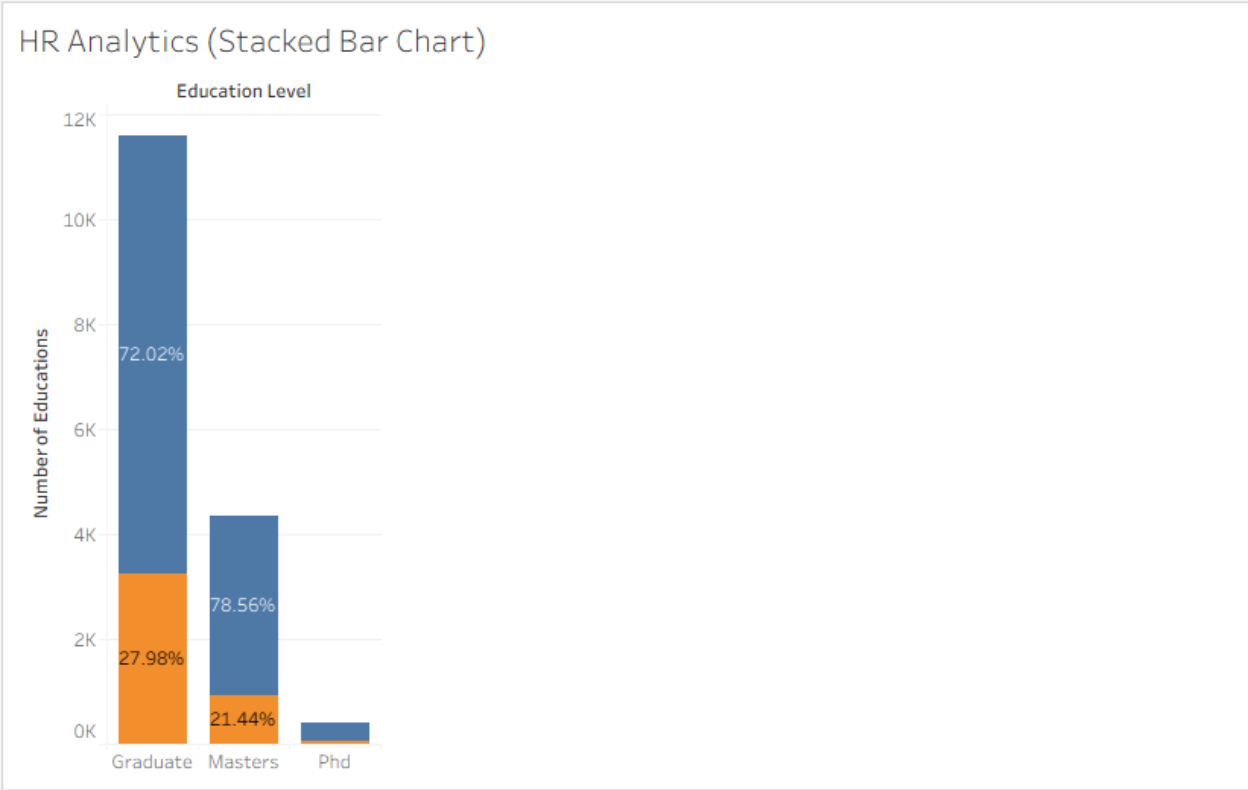
Entry 4: HR Turnover Analytics (Stacked Bar Chart)

1. **Name & ID:** Education Impact on Employee Turnover (HR-04)

2. **Visual Type:** Stacked Bar Chart with Percentages

3. **Source:** Kaggle (HR Analytics: Job Change of Data Scientists)

4. **Purpose:** To analyze how education levels influence a Data Scientist's intent to seek a new job.
5. **Message: Graduates** are the most active job seekers, while **PhD holders** show the highest retention and stability within the industry.
6. **Audience:** HR Directors and Talent Retention Teams.
7. **Design Choices:** I used a **Stacked Bar Chart** to show both the volume and the proportion of job seekers (**Concise**). Adding **Percent of Total** labels made the comparison between groups **Clear** and actionable.

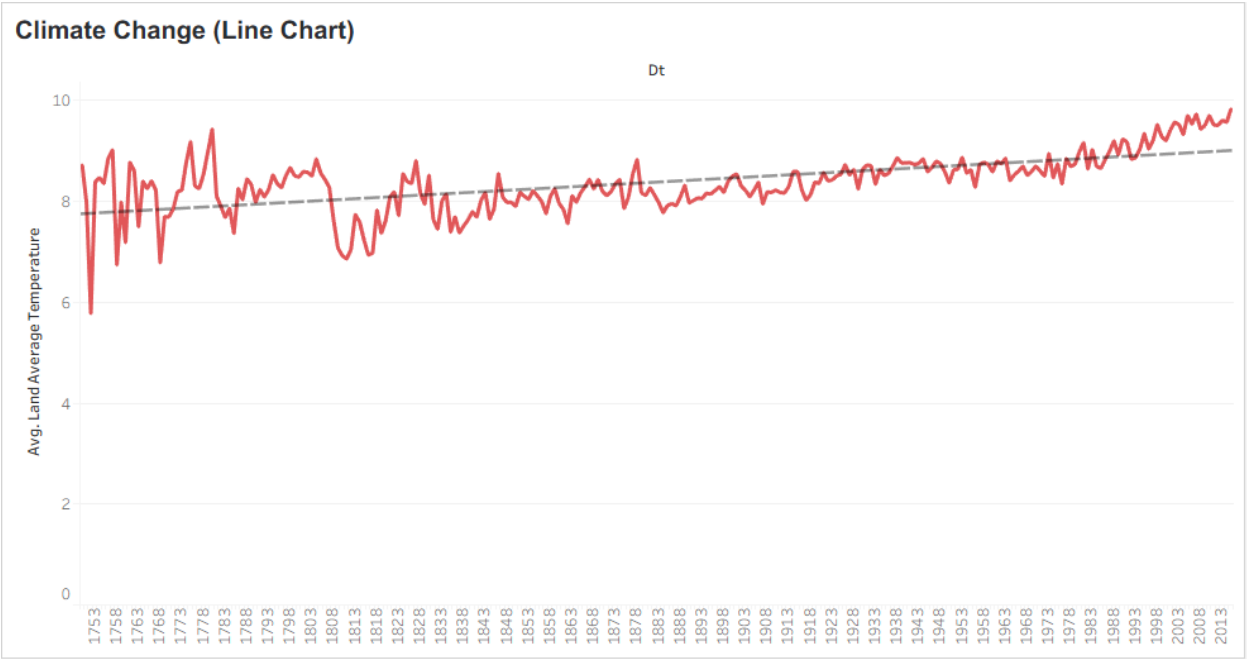


8. **Interactivity:** A filter for **Experience Level** (Newbie vs. Senior) would provide deeper insights into the turnover.
9. **Impact & Proof:** My peer immediately noticed the high orange segment (intent to leave) in the Graduate bar, confirming that the "Risk Group" was easily identifiable.

Entry 5: Global Warming Trends (Line Chart)

1. **Name & ID:** 250-Year Temperature Journey (CLIM-05)
2. **Visual Type:** Time-Series Line Chart
3. **Source:** Kaggle (Berkeley Earth Surface Temperature)
4. **Purpose:** To track the accelerating rise in Earth’s surface temperature since 1750.

5. **Message:** Global temperatures have risen sharply and consistently since the Industrial Revolution, with an alarming acceleration after 1950.
6. **Audience:** Environmental Policy Makers and Climate Scientists.
7. **Design Choices:** A **Line Chart** was used as it is the most effective tool for **Time-Series Analysis**. I chose a **Warm Red** color to emphasize the concept of 'warming' (**Captivating**). A **Trend Line** was added for long-term clarity.

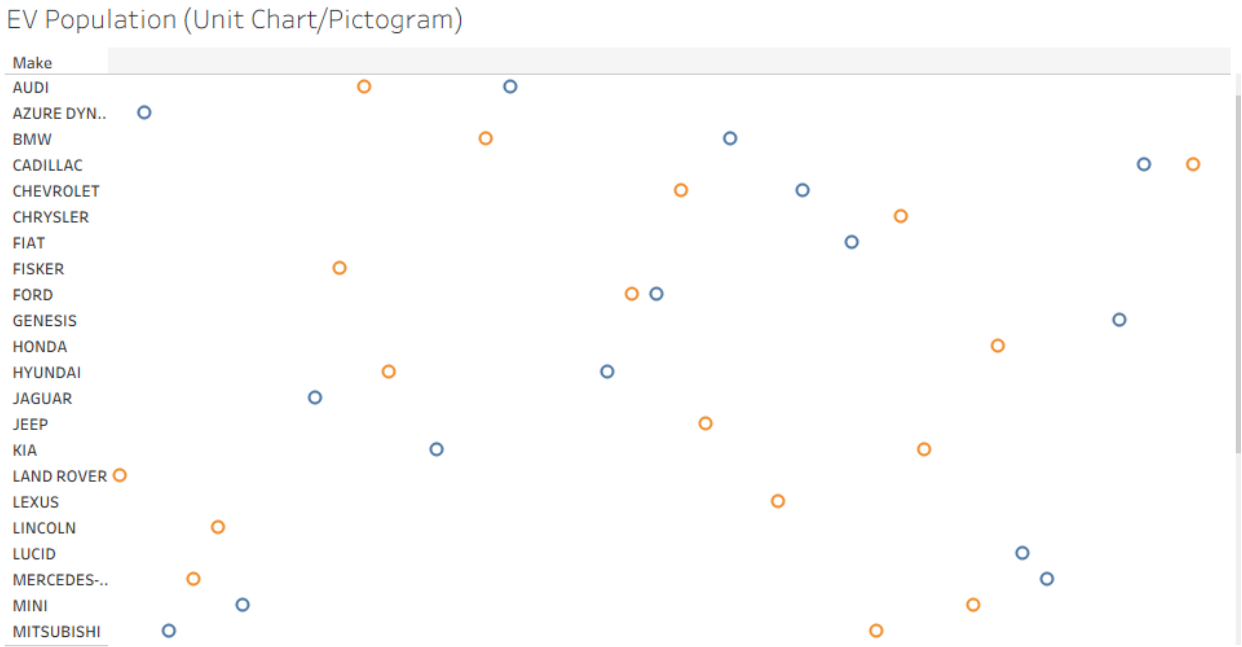


8. **Interactivity:** A **Zoom Feature** would allow users to look at specific centuries or decades in detail.
9. **Impact & Proof:** A peer noted the "hockey-stick" curve at the end of the timeline, proving that the message of accelerating climate change was successfully delivered.

Entry 6: EV Market Adoption (Unit Chart)

1. **Name & ID:** EV Manufacturer Unit Analysis (EV-06)
2. **Visual Type:** Unit Chart / Pictogram
3. **Source:** Kaggle (Electric Vehicle Population Data)
4. **Purpose:** To demonstrate manufacturer dominance in the EV sector using an intuitive, shape-based visual.
5. **Message: Tesla** dominates the market, and there is a significant consumer preference for Battery Electric Vehicles (BEV) over Hybrids.
6. **Audience:** Automotive Investors and Green Energy Advocates.

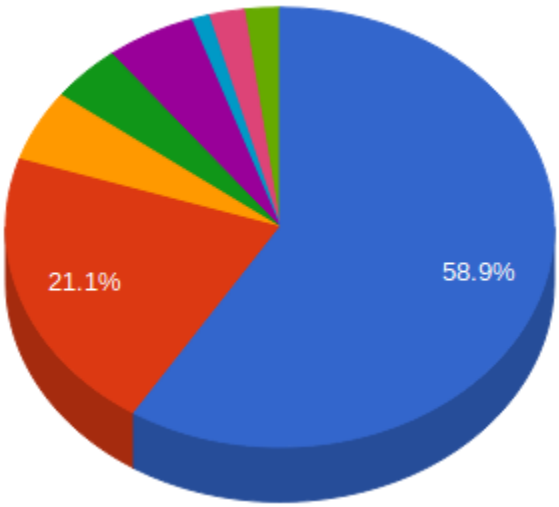
7. **Design Choices:** Fulfilling the coursework requirement for a **Pictogram**, I used repetitive **Circle/Car shapes** to represent volume. This makes the data feel more "tangible" and **Captivating** compared to a solid bar.



8. **Interactivity:** A **City Filter** was used to keep the visual **Clean** and avoid overcrowding the screen.

9. **Impact & Proof:** Peer review confirmed that the unit chart felt more like an "Infographic," making the huge volume differences between brands easier to digest.

Task D: Critique of a "Bad" Visualisation



Analysis:
This visualization is fundamentally flawed for the following four reasons:

1. **Violation of the 100% Rule:** The segments of the chart (if it's a pie chart) add up to more than 100%, which misleads the audience about the "part-to-whole" relationship.
2. **3D Distortion:** The use of 3D effects tilts the perspective, making the front segments look disproportionately larger than those in the back, violating the principle of **Accuracy**.
3. **Color Confusion:** The color palette lacks contrast, making it difficult for the viewer to distinguish between different categories at a glance.
4. **Information Overload (Clutter):** The chart contains too many labels and gridlines, failing the **Concise** criteria and creating cognitive load for the user.

References (APA 7th Edition)

1. Berkeley Earth. (2015). *Earth Surface Temperature Data*. Kaggle. [Link]
2. Gallup World Poll. (2019). *World Happiness Report*. Sustainable Development Solutions Network. [Link]
3. Shivamb, S. (2021). *Netflix Movies and TV Shows*. Kaggle. [Link]
4. Sootersaalu. (2019). *Amazon Top 50 Bestselling Books*. Kaggle. [Link]
5. Arashnic. (2020). *HR Analytics: Job Change of Data Scientists*. Kaggle. [Link]
6. Washington State DOL. (2024). *Electric Vehicle Population Data*. Kaggle. [Link]